```java
// Main.java
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        int[] n = {5, 4, 3, 2, 1};
        int n1 = n.length;

        MergeSort.mergeSort(n, 0, n1 - 1);
        System.out.println(Arrays.toString(n));
    }
}
```

```java
// MergeSort.java
public class MergeSort {
    public static void mergeSort(int[] arr, int low, int high) {
        if (low == high) return;

        int mid = (low + high) / 2;
        mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);

        merge(arr, low, mid, high);
    }

    public static void merge(int[] arr, int low, int mid, int high) {
        int left = low;
        int right = mid + 1;
        int[] temp = new int[high - low + 1];
        int k = 0;

        while (left <= mid && right <= high) {
            if (arr[left] < arr[right]) {
                temp[k++] = arr[left++];
            } else {
                temp[k++] = arr[right++];
            }
        }

        while (left <= mid) {
            temp[k++] = arr[left++];
        }

        while (right <= high) {
            temp[k++] = arr[right++];
        }

        for (int i = low; i <= high; i++) {
            arr[i] = temp[i - low];
        }
    }
}
```